

COMPASS_Synop- tic_LE_Tree_CH4_Flux_Calcs_2022

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1 Add Required Packages

2 Pull in Raw Datafiles

```
#defining where the raw data is

DATA_DIR_ROOT <- here::here(
  "Tree_Fluxes_Synoptic_LE",
  "2022",
  "Raw Data"
)

#function to read data
now <- function() format(Sys.time(), "%Y-%m-%d %H:%M:%S")
message(now(), " | Script started")
#finding the zip files
message(now(), " | Looking for ZIP files in: ", DATA_DIR_ROOT)

zip_files <- list.files(
  DATA_DIR_ROOT,
  pattern = "\\\\.zip$",
  full.names = TRUE)

message(now(), " | ZIP files found: ", length(zip_files))
```

```

print(zip_files)
  if (length(zip_files) == 0) {
    stop("No ZIP files found. Check DATA_DIR_ROOT.")}

cache <- list()

# Reader function
read_li7810_zip <- function(zip_file, tz = "US/Eastern") {
  message(now(), " | Processing ZIP: ", basename(zip_file))
  if (zip_file %in% names(cache)) {
    message(" -> Using cached version")
    return(cache[[zip_file]]) }

  tmp_dir <- tempfile("li7810_")
  dir.create(tmp_dir)
  on.exit(unlink(tmp_dir, recursive = TRUE), add = TRUE)
  unzip(zip_file, exdir = tmp_dir)
  data_files <- list.files(
    tmp_dir,
    pattern = "\\\\.data$",
    full.names = TRUE,
    recursive = TRUE )

  message(" -> .data files found: ", length(data_files))
  if (length(data_files) == 0) {
    stop("No .data files inside ZIP: ", zip_file) }

  dat_list <- lapply(data_files, function(f) {
    message("      Reading: ", basename(f))
    ffi_read_LI7810(f)})

  dat <- data.table::rbindlist(dat_list, fill = TRUE) %>%
    mutate(TIMESTAMP = with_tz(TIMESTAMP, tz))
  cache[[zip_file]] <- dat
  message(" -> ZIP finished: ", basename(zip_file))
  return(dat)}

# ACTUAL EXECUTION
message(now(), " | Starting data read")
alldat_list <- lapply(zip_files, read_li7810_zip)
alldat <- data.table::rbindlist(alldat_list, fill = TRUE)
message(now(), " | DONE")

```

```

message("Final number of rows: ", nrow(alldat))

# Show result
print(head(alldat))

#REMOVING WEIRD VERY LOW VALUES ON "2022-08-17 UP 7 148" cH4=-8000 licor bug???
alldat<-alldat%>%filter(CH4>-1000)

```

3 Read in metadata file and match to raw data

```

md <- "Raw Data/COMPASS_SynopticLE_TreeFlux_Metadata_2022b.csv"
dat <- read.csv(md)

metadat <- dat %>%
  # Remove rows without date
  filter(!is.na(Date)) %>%
  # Keep only desired instruments
  filter(SN %in% c("TG10-01448", "TG10-01426", "TG10-01028")) %>%
  # Normalize Plot labels
  mutate(
    Plot_raw = toupper(Plot),
    # Extract plot components
    #Site = str_extract(Plot_raw, "[A-Z]{3}"),
    #Zone = str_extract(Plot_raw, "(TR|UP)"), # somehow all of the UP PTR measurements turned
    Site = toupper(Site),
    Zone = toupper(Zone),
    Replicate = str_extract(Plot_raw, "(?<=TR|UP)[0-9]+"),
    # Letter ONLY if it comes AFTER a number
    Letter = str_extract(Plot_raw, "(?<=[0-9])[A-Z]$"),
    # Time normalization
    Time_start = str_replace_all(as.character(Time_start), "\\.", ":") |> str_trim(),
    Stop.time = str_replace_all(as.character(Stop.time), "\\.", ":") |> str_trim(),
    # Safe time parsing (NO WARNINGS)
    t_start = suppressWarnings(parse_date_time(Time_start, orders = c("H:M:S", "H:M"))),
    t_stop = suppressWarnings(parse_date_time(Stop.time, orders = c("H:M:S", "H:M"))),
    # Observation length
    Obs_length = as.numeric(t_stop - t_start, units = "secs"),
    # Timestamp
    TIMESTAMP = as.POSIXct(

```

```

    paste(Date, Time_start),
    format = "%m/%d/%Y %H:%M:%S",
    tz = "EST"
  ),
  # Sequential ID
  Seq = row_number(),
  # INAL standardized Plot label
  Plot = if_else(
    is.na(Letter),
    paste(Date, Site, Zone, Replicate, Seq, sep = "_"),
    paste(Date, Site, Zone, Replicate, Letter, Seq, sep = "_")) )

# Quick check
head(metadat$Plot)

```

4 Check start times overlaps or missing values

```

#checking if there is any start and stop time overlap from the field notes
overlaps_detected <- metadat %>%
  mutate(
    start_ts = as.POSIXct(
      paste(Date, Time_start),
      format = "%m/%d/%Y %H:%M:%S",
      tz = "EST"
    ),
    end_ts = start_ts + Obs_length
  ) %>%
  arrange(SN, start_ts) %>%
  group_by(SN) %>%
  mutate(
    next_plot      = lead(Plot),
    next_start_ts  = lead(start_ts),
    overlap_fffi   = !is.na(next_start_ts) & next_start_ts <= end_ts
  ) %>%
  filter(overlap_fffi) %>%
  select(
    SN,
    Plot,
    start_ts,

```

```

    end_ts,
    next_plot,
    next_start_ts,
    Obs_length
  )

print(overlaps_detected)

#checking problems with obs_length - this is a frequent bug for manual field notes
field_notes_overlap<-metadat %>%
  mutate(
    obs_problem =
      is.na(Obs_length) | Obs_length <= 0
  ) %>%
  filter(obs_problem) %>%
  select(
    SN,
    Date,
    Plot,
    Time_start,
    Stop.time,
    Obs_length
  )

print(field_notes_overlap)

# checking unique measurment start time and stop time
# alldat %>%
#   filter(
#     SN == "TG10-01448",
#     TIMESTAMP > as.POSIXct("2023-01-05 13:28:00"),
#     TIMESTAMP < as.POSIXct("2023-01-05 13:33:00")
#   ) %>%
#   ggplot(aes(x = TIMESTAMP, y = CO2)) +
#   geom_point()#ylim(300,500)

```

5 Match metadata to raw data

```

#function to match data per equipment (considering different serial numbers) uhul!
match_flux_metadata <- function(alldat, metadat) {

```

```

message("Starting metadata  Licor matching")
#Sanity checks (fail early)
stopifnot(
  "TIMESTAMP" %in% names(alldat),
  "SN"         %in% names(alldat),
  "Date"       %in% names(metadat),
  "Time_start"%in% names(metadat),
  "Obs_length"%in% names(metadat),
  "Plot"       %in% names(metadat)
)

stopifnot(all(metadat$Obs_length > 0, na.rm = TRUE))
stopifnot(anyDuplicated(metadat$Plot) == 0)
# Instruments present in BOTH datasets
instruments <- intersect(
  unique(alldat$SN),
  unique(metadat$SN)
)
message("Instruments to process: ", paste(instruments, collapse = ", "))
results <- lapply(instruments, function(sn) {
  message("Processing SN: ", sn)
  dat_i <- alldat %>%
    filter(SN == sn)
  meta_i <- metadat %>%
    filter(SN == sn) %>%
    arrange(Date, Time_start) %>%
    mutate(metadat_row = row_number())
  stopifnot(nrow(meta_i) > 0)
# Actual matching
dat_i$metadat_row <- ffi_metadata_match(
  data_timestamps = dat_i$TIMESTAMP,
  start_dates     = meta_i$Date,
  start_times     = meta_i$Time_start,
  obs_lengths     = meta_i$Obs_length )
# Attach metadata (vectorized, fast)
dat_i <- dat_i %>%
  mutate(
    Plot      = meta_i$Plot[metadat_row],
    Volume    = meta_i$Volume[metadat_row],
    Temp      = meta_i$Temp[metadat_row],
    Area      = meta_i$Area[metadat_row],
    Dead_band = meta_i$Dead_band[metadat_row]
  )

```

```

# Drop unmatched rows
n_before <- nrow(dat_i)
dat_i <- dat_i %>% filter(!is.na(metadata_row))
n_after <- nrow(dat_i)
message("  Matched rows: ", n_after, " / ", n_before)
dat_i })
alldat_matched <- rbindlist(results, fill = TRUE)
#Final sanity checks
stopifnot(
  !anyNA(alldat_matched$Plot),
  nrow(alldat_matched) > 0
)
message("Matching finished successfully")
return(alldat_matched)}

#running the function
message("Running final metadata matching...")
alldat_matched <- match_flux_metadata(alldat, metadata)
message("Matched dataset rows: ", nrow(alldat_matched))
head(alldat_matched)

#saving matched data and metadata as raw data is too big (today, Licor files leave in the CO2
saveRDS(alldat_matched, file = "alldat_matched_2022.rds")

#can remove any row that have NA's in the CH4 column
#alldat <- alldat %>%
# filter(!is.na(CH4))

```

6 reating alldata matched df

```

# after you match the data for the first time - we can just read the .rds file to save process
# if you need to match licor data and metadata again, remove "#" from the previous chunks

alldat_matched <- readRDS("alldat_matched_2022.rds")

```

```

#Extracting initial CH4 concentration

```

```

## Extract initial CH4 concentration (t0) for each Plot for QAQC later
#saving df to use after flux calculation

```

```
df_CH4_in <- alldat_matched %>%
  arrange(Plot, TIMESTAMP) %>%
  group_by(Plot) %>%
  summarise(CH4_initial = first(CH4), .groups = "drop")
```

7 Creating temperature csv from the L2-weather station in the wetland

```
# Getting Air temp from the weather station
## Read L2 temperature data from ZIP stored outside GitHub,
## extract desired variable, save lightweight CSV,
## and remove temporary unzipped files afterwards

## Read extracted L2 Parquet files and save temperature time series
BASE_DIR <- "D:/COMPASS_L2-v2-1/v2-1"
SITE      <- "(OWC|CRC|PTR)"
VARIABLE  <- "wx-tempavg15"

OUTFILE   <- "Raw Data/L2_temp_timeseries.csv"

# Build regex that matches REAL L2 filenames
regex <- paste0(SITE, ".*", VARIABLE, ".*_L2_v2-1.*\\.parquet$")

# Locate Parquet files
files <- list.files(BASE_DIR, pattern = regex, recursive = TRUE, full.names = TRUE)

message("Files found: ", length(files))
print(basename(files))

# Fail fast if nothing is found
stopifnot(length(files) > 0)

# Read and combine Parquet files
dat <- lapply(files, function(f) {
  message("Reading ", basename(f))
  read_parquet(f)
})

dat <- bind_rows(dat)
```

```

# Optional QA filtering (only if column exists)
if ("QA_flag" %in% names(dat)) {
  dat <- dat %>% filter(QA_flag == 0)
}

# Keep only necessary columns
temps <- dat %>%
  select(TIMESTAMP, Plot, Site, Value) %>%
  rename(Temp = Value)

head(temps)

# Save lightweight output only
write.csv(
  temps,
  file = OUTFILE,
  row.names = FALSE
)

```

8 Reading Temp df to merge to metadat

```

#reading temp
temps <- read.csv("Raw Data/L2_temp_timeseries.csv")

## Match up the L1 temp data and add it to the metadata dataframe
#Going to need to use site and timestamp in the metadata in order to find a date

#make sure time stamps are in the correct format
alldat_matched$TIMESTAMP <- as.POSIXct(alldat_matched$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")
#problem here L2 data for midnight is always "2022-01-01" we loose this line to NA when form
temps$TIMESTAMP[!grepl(":", temps$TIMESTAMP)] <-
  paste0(temps$TIMESTAMP[!grepl(":", temps$TIMESTAMP)], " 00:00:00")
temps$TIMESTAMP <- as.POSIXct(temps$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")

#time rounding step
alldat_matched$TIMESTAMP_round <- round_date(alldat_matched$TIMESTAMP, "60 mins")
temps$TIMESTAMP_round <- round_date(temps$TIMESTAMP, "60 mins")

#ensure there is only one data observation per hour

```

```
temps_clean <- temps %>%
  group_by(TIMESTAMP_round, Site) %>%
  summarize(Temp = mean(Temp ))
```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by TIMESTAMP_round and Site.
- i Output is grouped by TIMESTAMP_round.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(TIMESTAMP\_round, Site))` for per-operation grouping (`?dplyr::dplyr\_by`) instead.

```
#join them
alldat_matched <- alldat_matched %>%
  left_join(temps_clean, by=c("Site", "TIMESTAMP_round"))
```

9 fixing some Deadbands and stop time

```
# Table of plots with custom deadband adjustments
time_overrides <- tibble::tibble(
  Plot = c("10/21/2022_OWC_TR_121_56",
           "8/17/2022_OWC_UP_7_148",
           "10/7/2022_PTR_TR_7_93",
           "10/4/2022_CRC_UP_306_68"
  ),
  dead_band_manual = c(120,120,65,4),
  stop_offset_secs = c(0,0,0,0))

#apply new incubation period
alldat_trimmed <- alldat_matched %>%
  left_join(time_overrides, by = "Plot") %>%
  group_by(Plot) %>%
  mutate(
    t_start_original = min(TIMESTAMP),
    t_end_original   = max(TIMESTAMP),

    # deadband manual or metadat
    t_start_adjusted = if_else(
      !is.na(dead_band_manual),
      t_start_original + dead_band_manual,
```

```

    t_start_original + as.numeric(coalesce(first(Dead_band), 0))
  ),

  # new offset
  t_end_adjusted = if_else(
    !is.na(stop_offset_secs),
    t_end_original + stop_offset_secs,
    t_end_original
  )
) %>%
filter(TIMESTAMP >= t_start_adjusted,
       TIMESTAMP <= t_end_adjusted) %>%
ungroup() %>%
select(-t_start_original, -t_end_original,
       -t_start_adjusted, -t_end_adjusted,
       -dead_band_manual, -stop_offset_secs)

# alldat_trimmed %>%
#   filter(Plot == "8/17/2022_OWC_UP_6_147") %>%
#   ggplot(aes(TIMESTAMP, CH4)) +
#   geom_point() + theme_classic()

```

10 Calculate fluxes

```

# TO Do - NEED TO PULL TEMP from the weather station
#change the units
alldat_trimmed$CH4_nmol <- ffi_ppb_to_nmol(alldat_trimmed$CH4,
                                           volume = alldat_trimmed$Volume, # m3
                                           temp = alldat_trimmed$Temp) # degrees C

```

Assuming atm = 101325 Pa

Using R = 8.31446261815324 m3 Pa K-1 mol-1

```

#> Assuming atm = 101325 Pa
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1

#using chamber area here

```


[illegible]

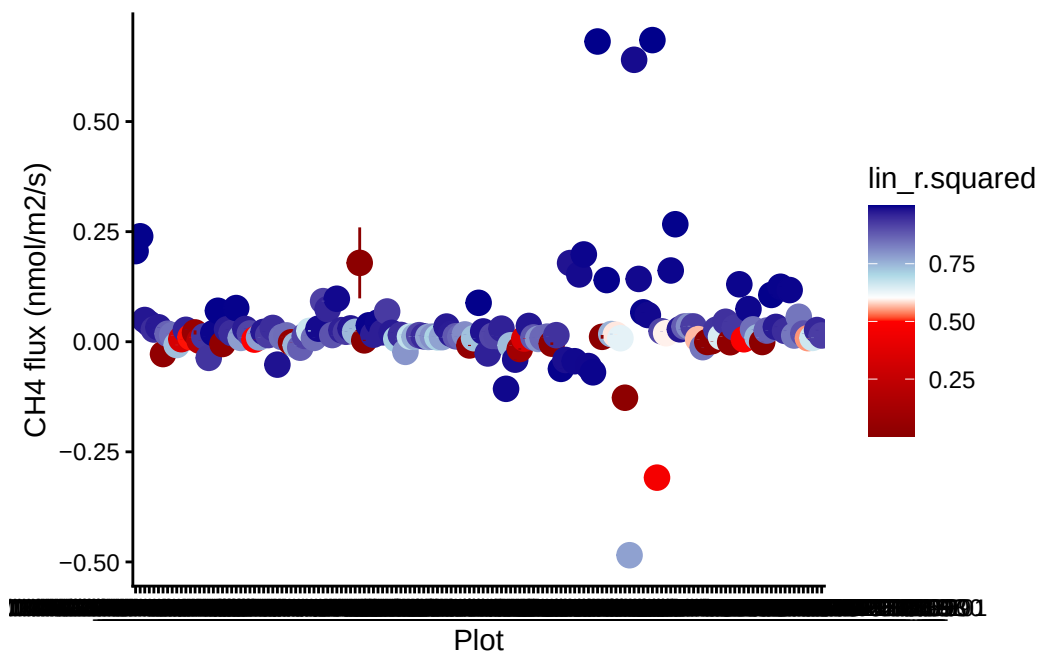
```
head(fluxes)
```

	Plot	TIMESTAMP	HM81_AIC	HM81_flux.estimate		
1	10/14/2022_CRC_TR_309_71	2022-10-14 10:44:29	3515.891	0.24120549		
2	10/14/2022_CRC_TR_311_72	2022-10-14 10:33:24	NA	NA		
3	10/14/2022_CRC_TR_312_73	2022-10-14 10:26:29	NA	NA		
4	10/14/2022_CRC_TR_313_74	2022-10-14 11:19:44	NA	NA		
5	10/14/2022_CRC_TR_314_75	2022-10-14 10:06:29	3027.708	0.04699297		
6	10/14/2022_CRC_TR_315_76	2022-10-14 10:13:19	1793.304	0.05172906		
	HM81_p.value	HM81_r.squared	HM81_RMSE	lin_AIC	lin_flux.estimate	
1	3.072741e-128	0.6603094	6.251464	804.2856	0.20581265	
2	NA	NA	NA	1142.5590	0.23942096	
3	NA	NA	NA	566.5954	0.04937976	
4	NA	NA	NA	1176.4514	0.04091208	
5	2.754526e-254	0.7254439	1.298098	1587.7168	0.02777137	
6	6.724009e-154	0.6892433	1.074817	379.9575	0.03284502	
	lin_flux.std.error	lin_int.estimate	lin_int.std.error	lin_p.value		
1	0.0004204426	5673.876	0.04351158	0		
2	0.0005016667	5598.863	0.05481383	0		
3	0.0003373849	5697.073	0.03491594	0		
4	0.0002249392	5570.595	0.03496713	0		
5	0.0002245043	5627.389	0.03878806	0		
6	0.0002340551	5614.331	0.02692498	0		
	lin_r.squared	lin_RMSE	poly_AIC	poly_r.squared	poly_RMSE	rob_AIC
1	0.9977598	0.5076668	254.6281	0.9991965	0.3046129	807.2760
2	0.9975124	0.6569157	849.8591	0.9985219	0.5072766	1145.0641
3	0.9755002	0.4073781	529.9348	0.9772772	0.3930573	566.6602
4	0.9761571	0.4989739	415.9243	0.9907222	0.3116443	1176.5882
5	0.9445675	0.5832761	773.7157	0.9776623	0.3706768	1588.4156
6	0.9705282	0.3309997	261.0747	0.9759862	0.2992832	380.6330
	rob_converged	rob_flux.estimate	rob_flux.std.error	rob_RMSE		
1	TRUE	0.20539950	0.0004306100	0.5263496		
2	TRUE	0.24006982	0.0005203044	0.7005841		
3	TRUE	0.04929934	0.0003279813	0.3973067		
4	TRUE	0.04097230	0.0002355315	0.5586385		
5	TRUE	0.02776161	0.0002348340	0.6394551		
6	TRUE	0.03266663	0.0002371702	0.3081249		
	TIMESTAMP_min	TIMESTAMP_max				
1	2022-10-14 10:43:00	2022-10-14 10:45:59				
2	2022-10-14 10:31:50	2022-10-14 10:34:59				
3	2022-10-14 10:25:00	2022-10-14 10:27:59				
4	2022-10-14 11:17:30	2022-10-14 11:21:59				

```
5 2022-10-14 10:04:00 2022-10-14 10:08:59
6 2022-10-14 10:11:40 2022-10-14 10:14:59
```

```
#merging fluxes with initial CH4 concentration
fluxes <- fluxes %>%
  left_join(df_CH4_in, by = "Plot")

#take a look at the fluxes we calculated with R2
ggplot(fluxes, aes(Plot, lin_flux.estimate, color = lin_r.squared)) +
  geom_point(size=4) + theme_classic() +
  geom_linerange(aes(ymin = lin_flux.estimate- lin_flux.std.error,
                    ymax = lin_flux.estimate+ lin_flux.std.error)) +
  scale_color_gradientn(colours = c("darkred", "red", "white", "lightblue", "darkblue"),
                       values = c(0, 0.5, 0.6, 0.7, 1)) +
  ylab("CH4 flux (nmol/m2/s)")
```



```
#sort fluxes by R2
fluxes_good <- as.data.frame(subset(fluxes, lin_r.squared >= .90, select = Plot:TIMESTAMP_ma
fluxes_lowr2 <- as.data.frame(subset(fluxes, lin_r.squared < .90 & lin_flux.estimate < 1, se
fluxes_weird <- as.data.frame(subset(fluxes, lin_r.squared < .90, select = Plot:TIMESTAMP_ma

#find fluxes that represent the first and fourth quartile of the data
```

```

low <- quantile(fluxes$lin_flux.estimate, 0.05)
high <- quantile(fluxes$lin_flux.estimate, 0.95)

fluxes_low <- as.data.frame(subset(fluxes, lin_flux.estimate <= low, select = Plot:TIMESTAMP))
fluxes_high <- as.data.frame(subset(fluxes, lin_flux.estimate >= high, select = Plot:TIMESTAMP))

tails <- rbind(fluxes_low, fluxes_high)

```

11 check the fluxes for which I changed the start time due to lags or high values (recorded in notes of metadata)

```

# grab trim limits for each overridden Plot
trim_limits <- alldat_trimmed %>%
  filter(Plot %in% time_overrides$Plot) %>%
  group_by(Plot) %>%
  summarise(
    t_start_trim = min(TIMESTAMP),
    t_end_trim   = max(TIMESTAMP) )

# use full alldat_matched + trim limits to classify each point
r_all <- alldat_matched %>%
  filter(Plot %in% time_overrides$Plot) %>%
  left_join(trim_limits, by = "Plot") %>%
  group_by(Plot) %>%
  mutate(
    time_sec = as.numeric(difftime(TIMESTAMP, min(TIMESTAMP), units = "secs")),
    point_status = case_when(
      TIMESTAMP < t_start_trim ~ "Discarded (deadband)",
      TIMESTAMP > t_end_trim   ~ "Discarded (stop trim)",
      TRUE                     ~ "Used" )) %>%
  ungroup()

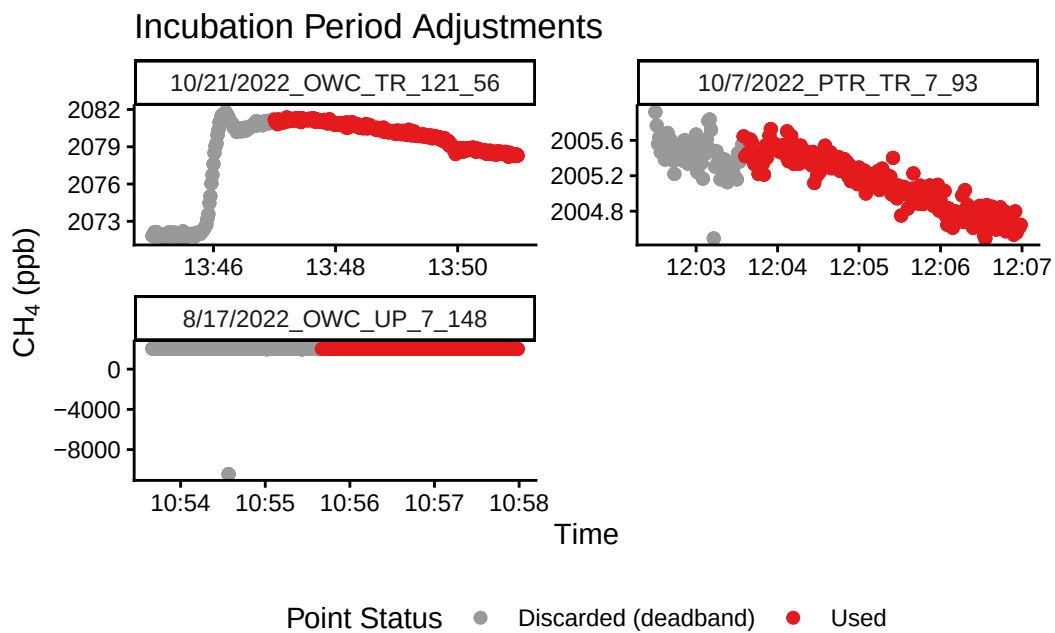
# plot all overridden plots
ggplot(r_all, aes(x = TIMESTAMP, y = CH4, color = point_status)) +
  geom_point(size = 1.8) +
  facet_wrap(~ Plot, scales = "free", ncol = 2) +
  scale_color_manual(
    values = c(
      "Used" = "#E41A1C",

```

```

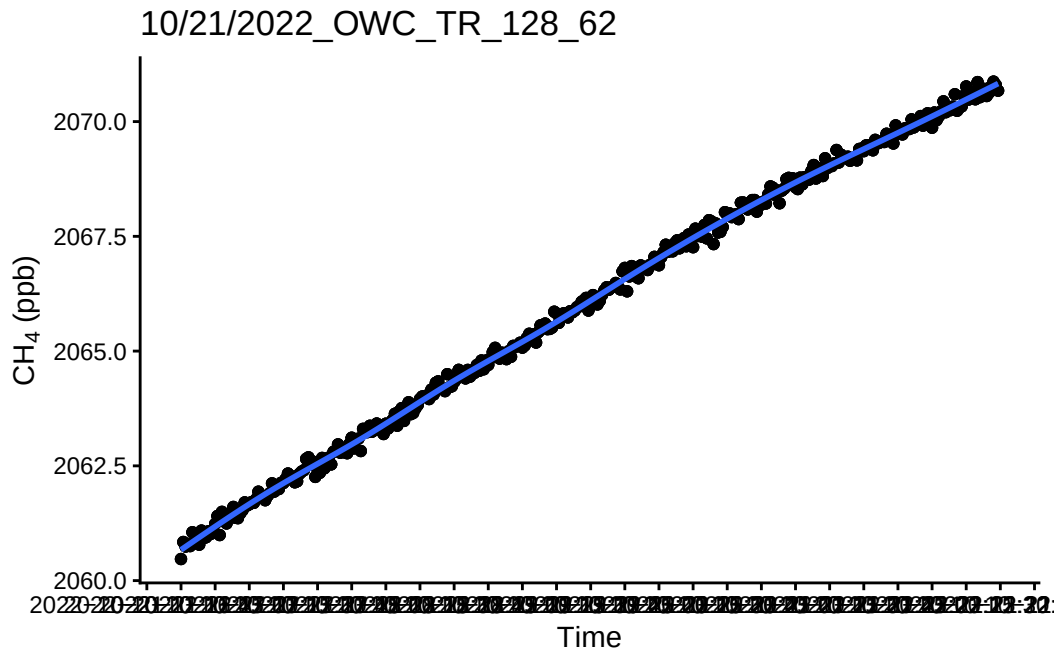
  "Discarded (deadband)" = "grey60",
  "Discarded (stop trim)" = "grey20"
)
) +
theme_classic() +
ylab(expression(paste(CH[4], " (ppb)"))) +
xlab("Time") +
labs(
  title = "Incubation Period Adjustments",
  color = "Point Status") +
theme(legend.position = "bottom")

```

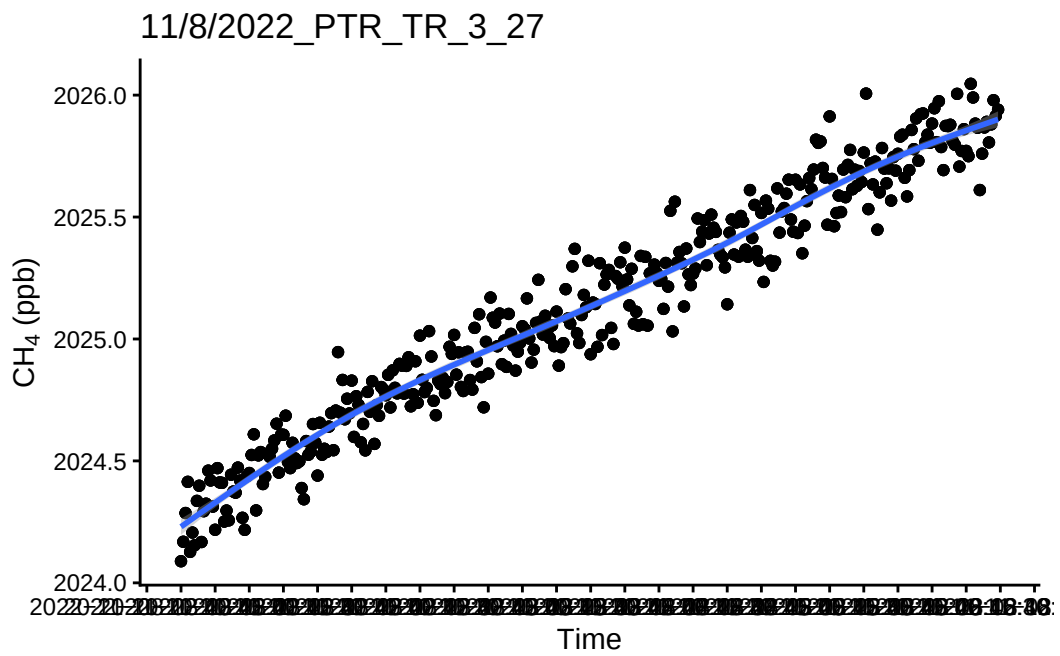


12 Spot check a few fluxes with $R^2 > 0.9$

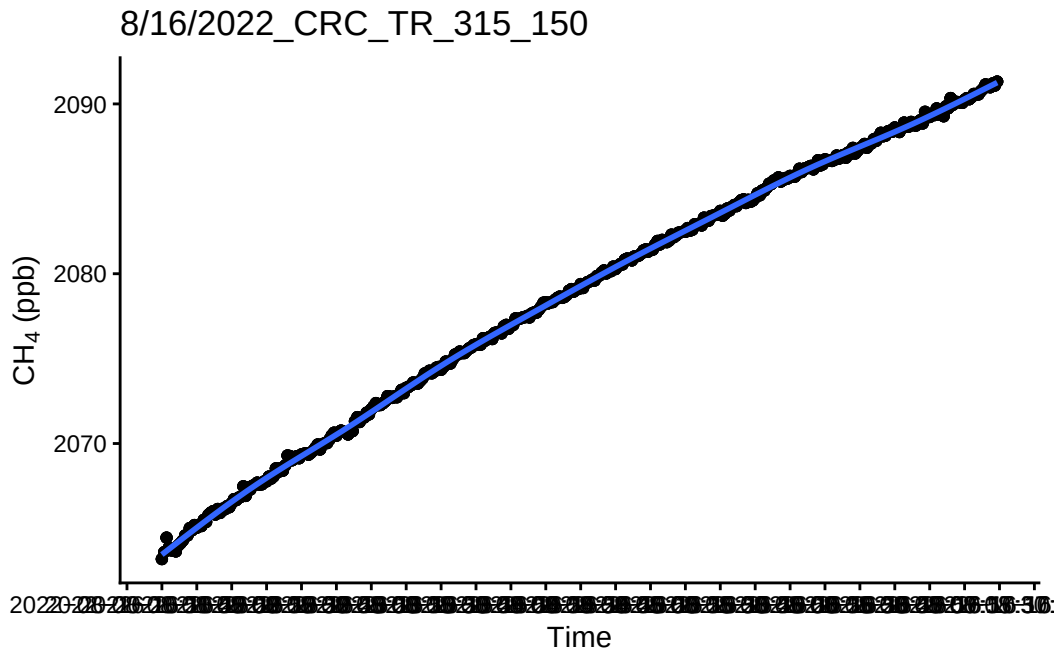
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



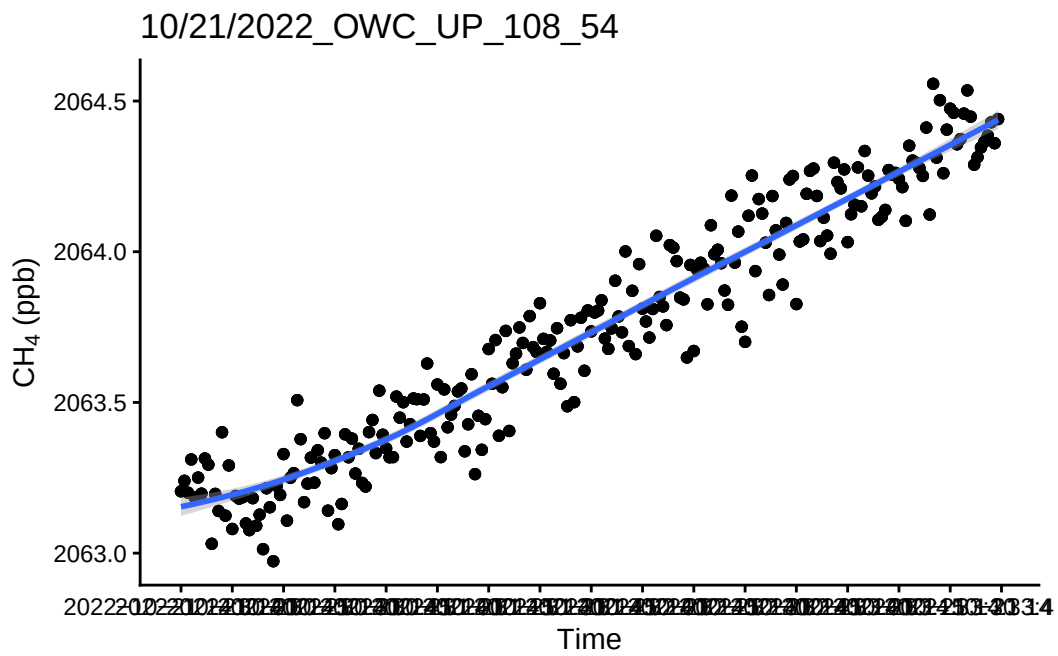
``geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'`



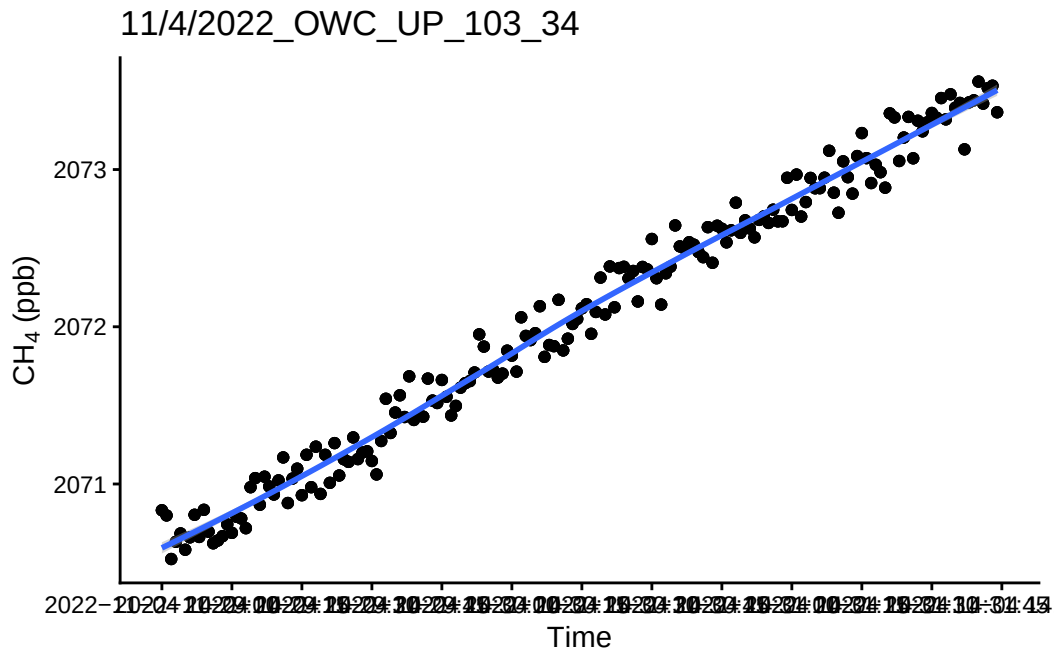
``geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'`



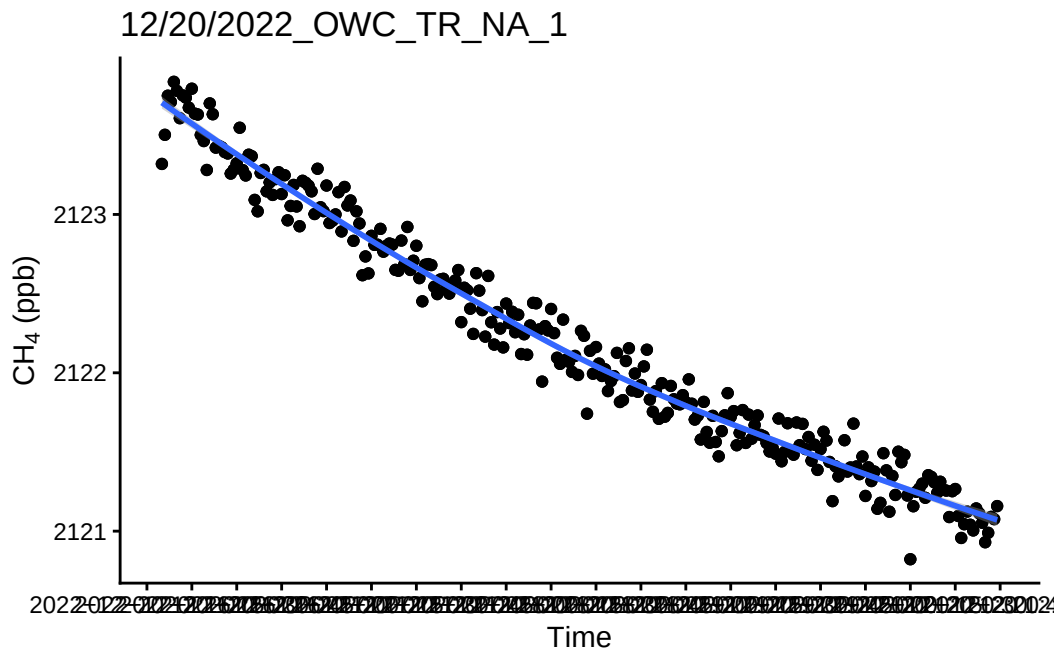
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



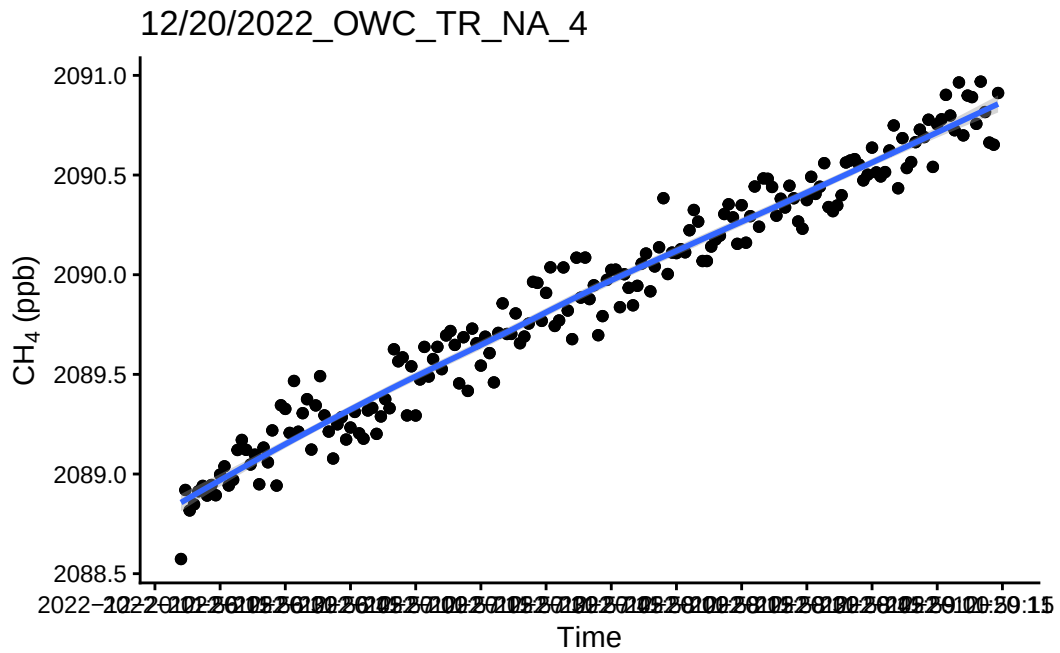
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



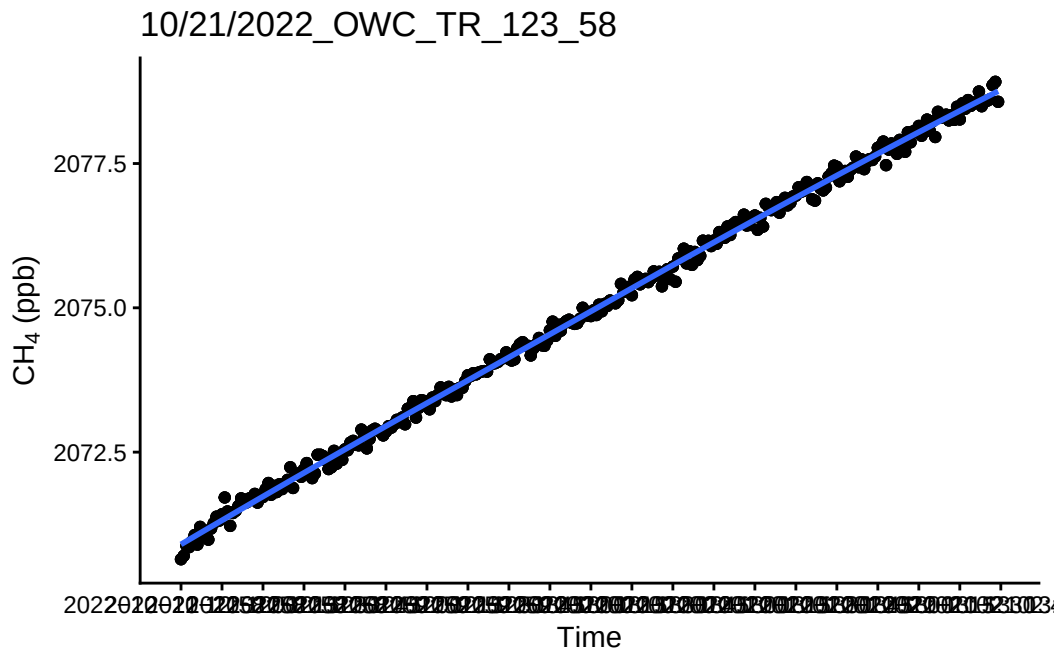
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



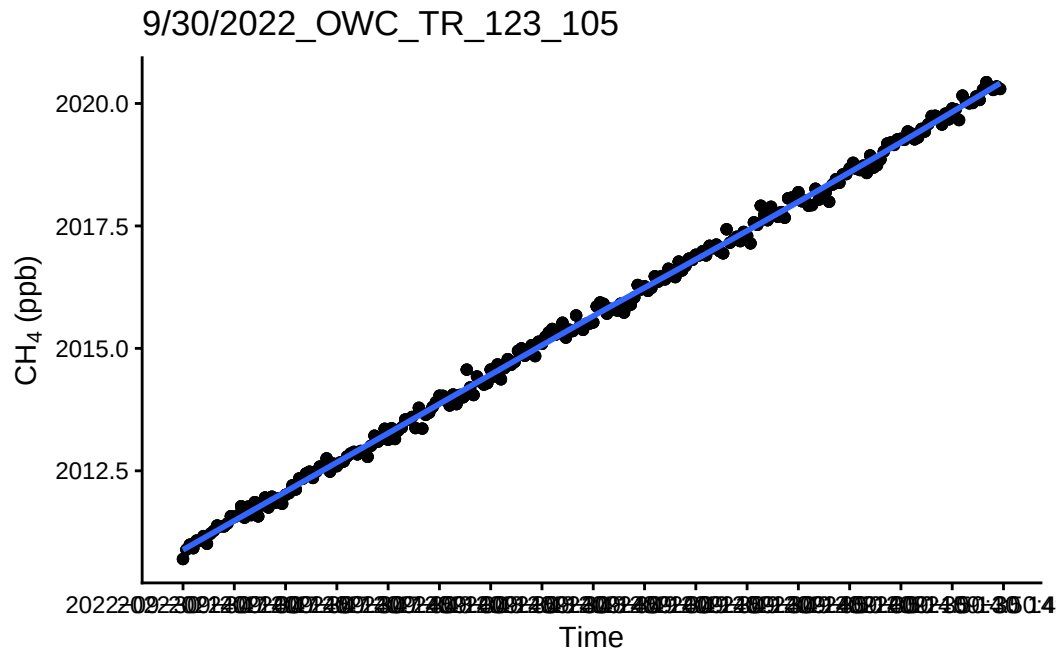
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

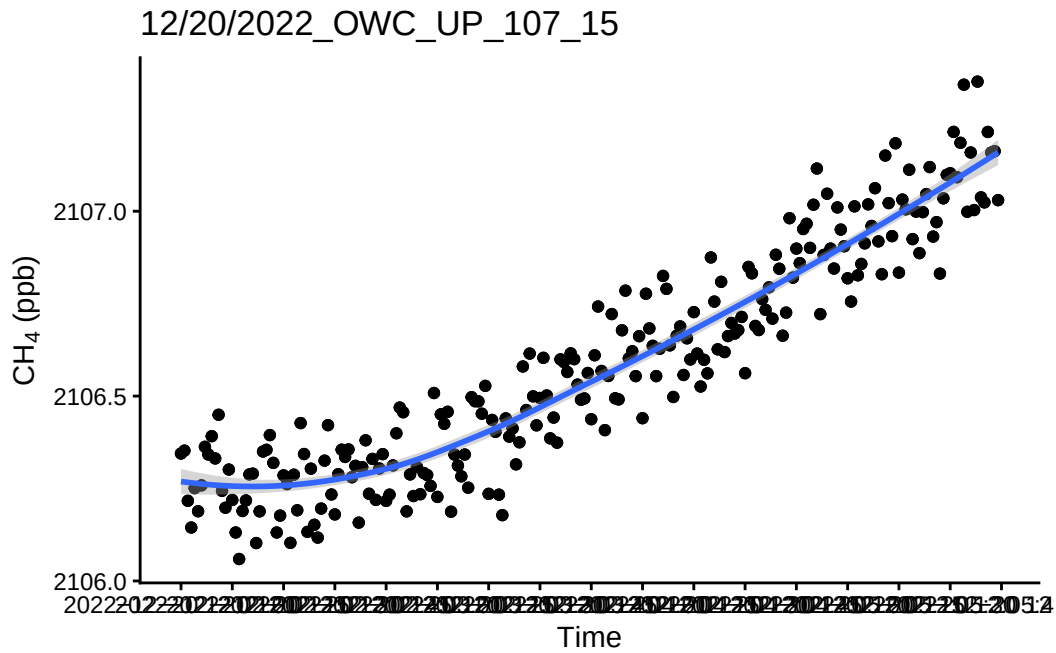


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

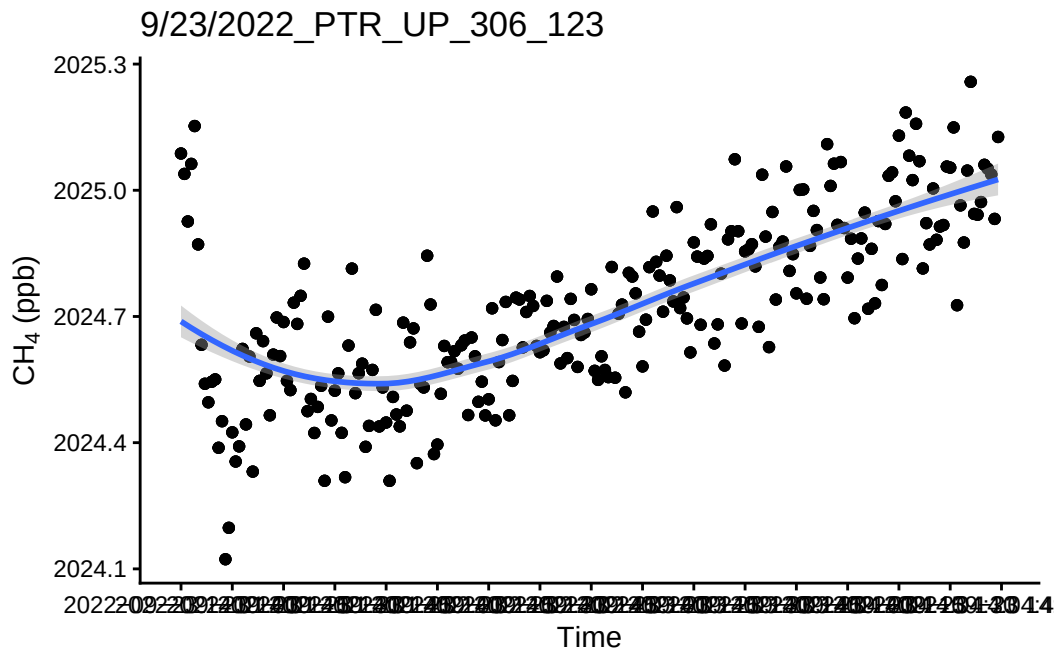


13 Take a look at the fluxes < 0.9 and a flux estimate below 1

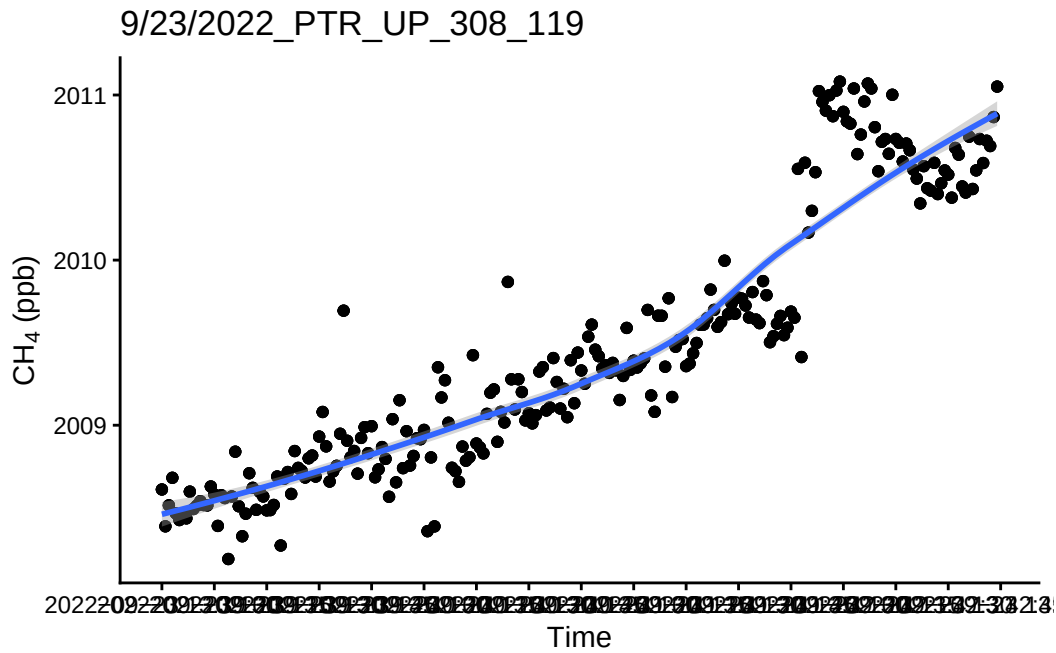
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



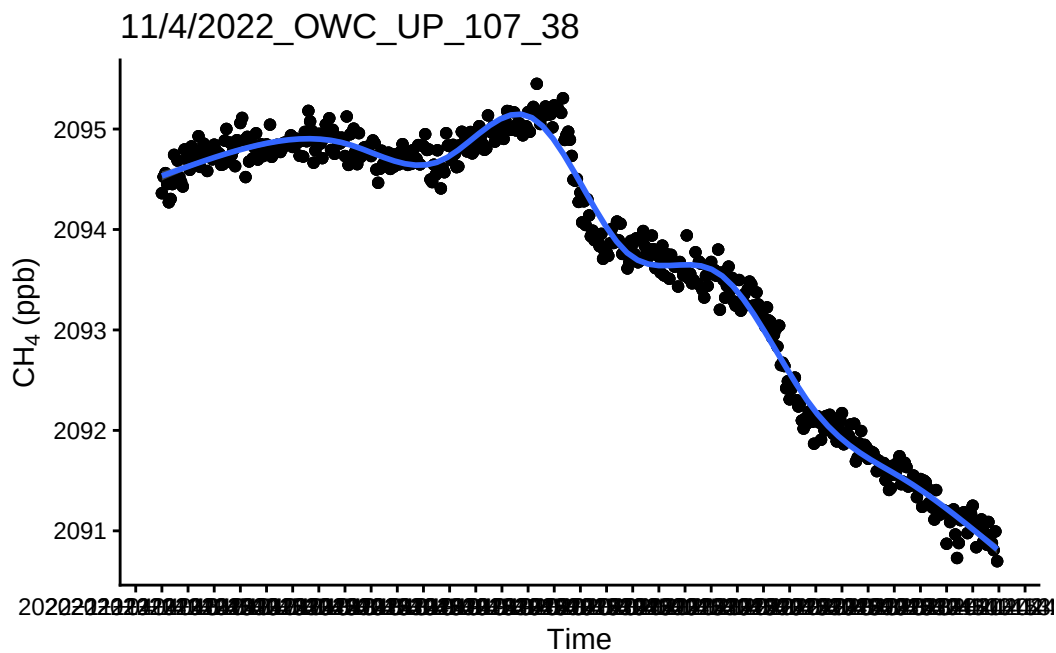
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



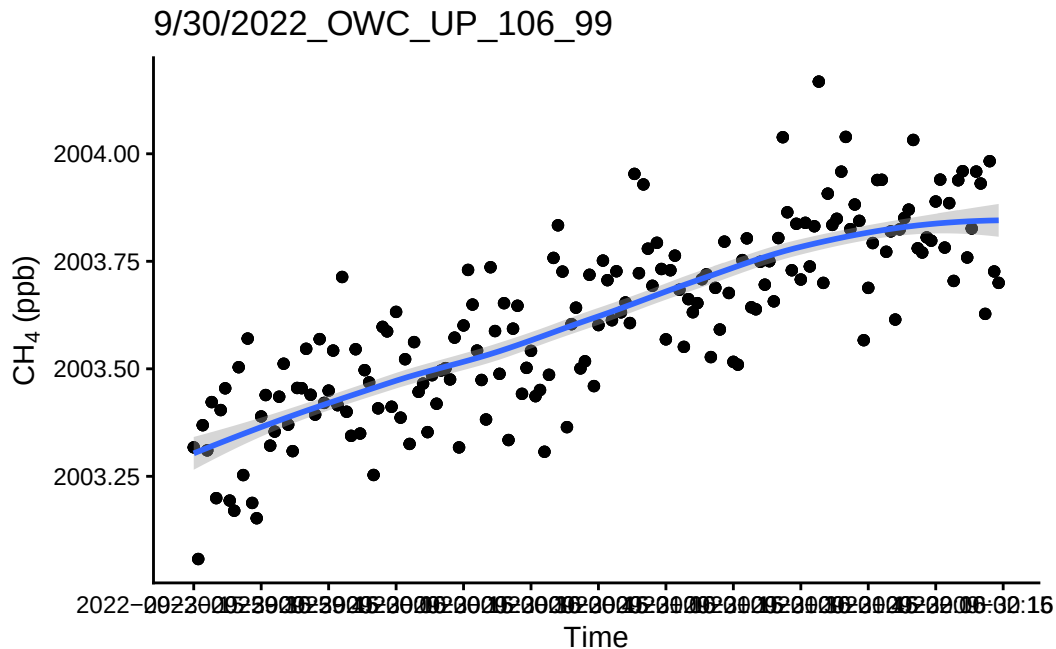
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



14 Take a look at the bad fluxes <0.9 and have a high flux estimate

15 See if we want to remove these fluxes or if we want to try and recalculate these ones?

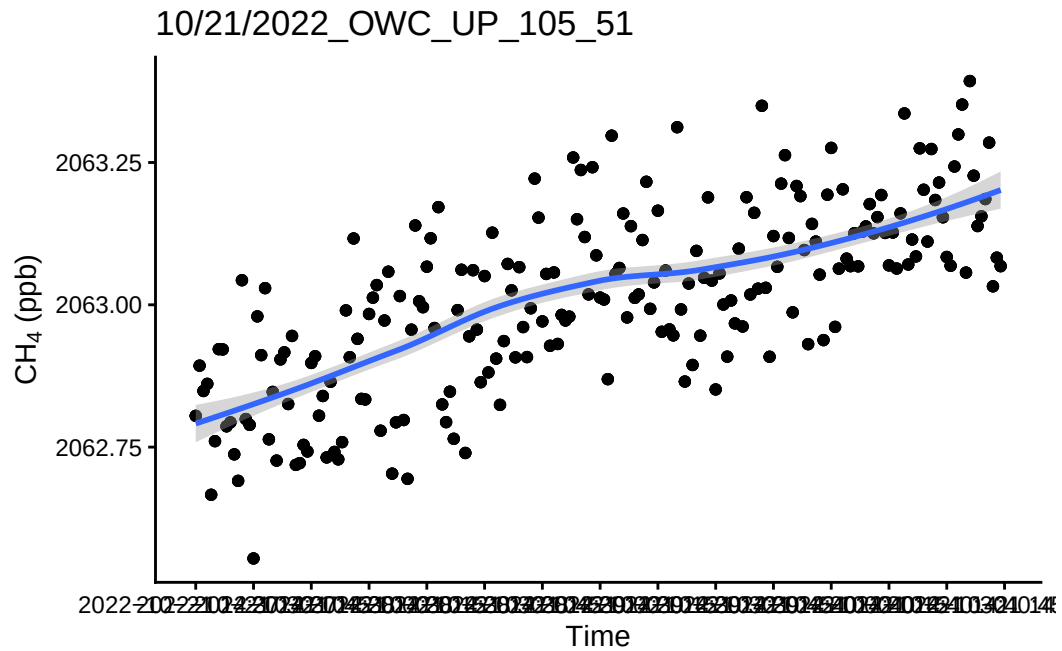
```
### pull out fluxes with R2 < 0.9 and put them in a list
bad_plots <- unique(fluxes_weird$Plot)

# take a random sample (FALSE so we don't get the same plot twice)
# set.seed(...) to guarantee reproducibility
bad_sample <- sample(bad_plots, 5, replace=FALSE)

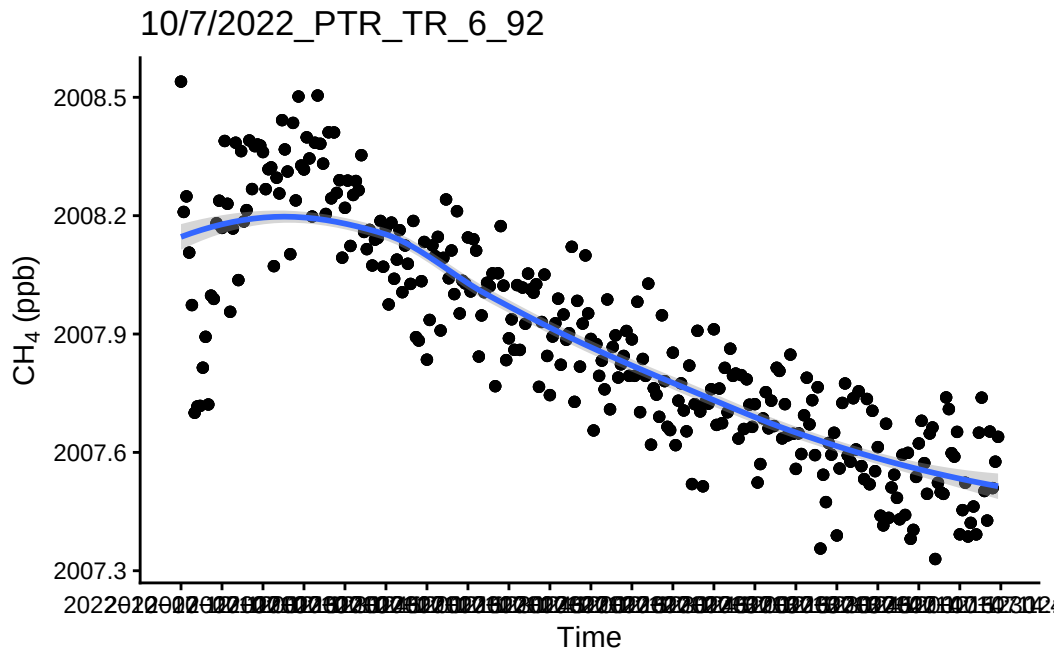
#Loop to run five random plots subset above
for(P in bad_sample){
  r1 <- subset(alldat_trimmed, Plot == P & !is.na(metadata_row))
  p1 <- ggplot(r1, aes(TIMESTAMP, CH4)) + geom_point() + theme_classic() +
    scale_x_datetime(breaks = "15 sec") +
    ylab(expression(paste( CH [4], " (ppb)"))) +
    xlab("Time") + labs(title = P) +
```

```
geom_smooth()
print(p1)
}
```

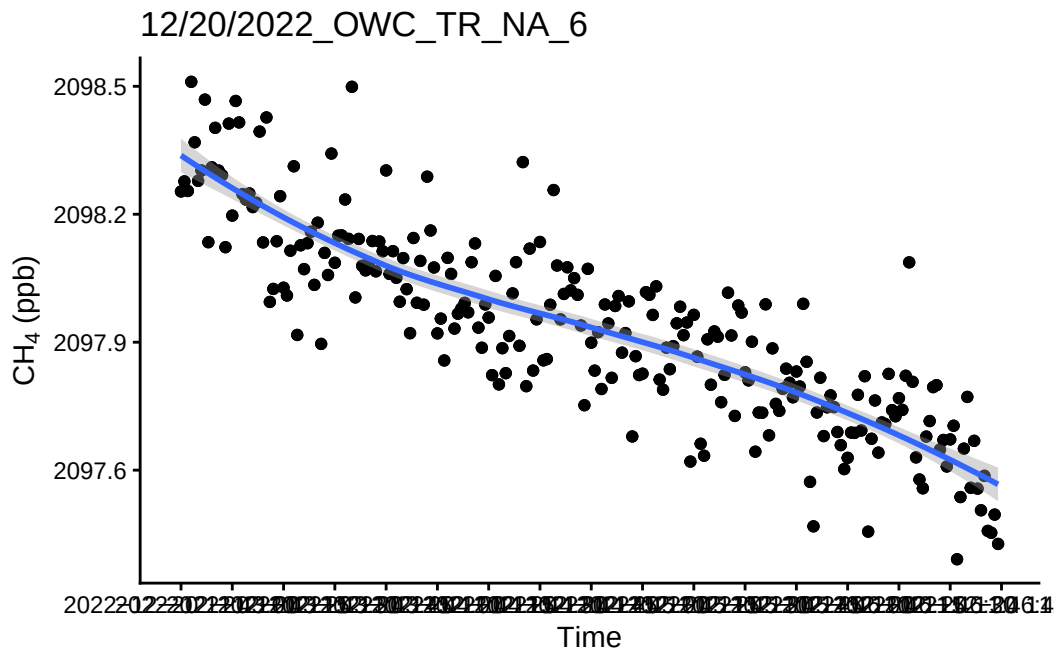
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



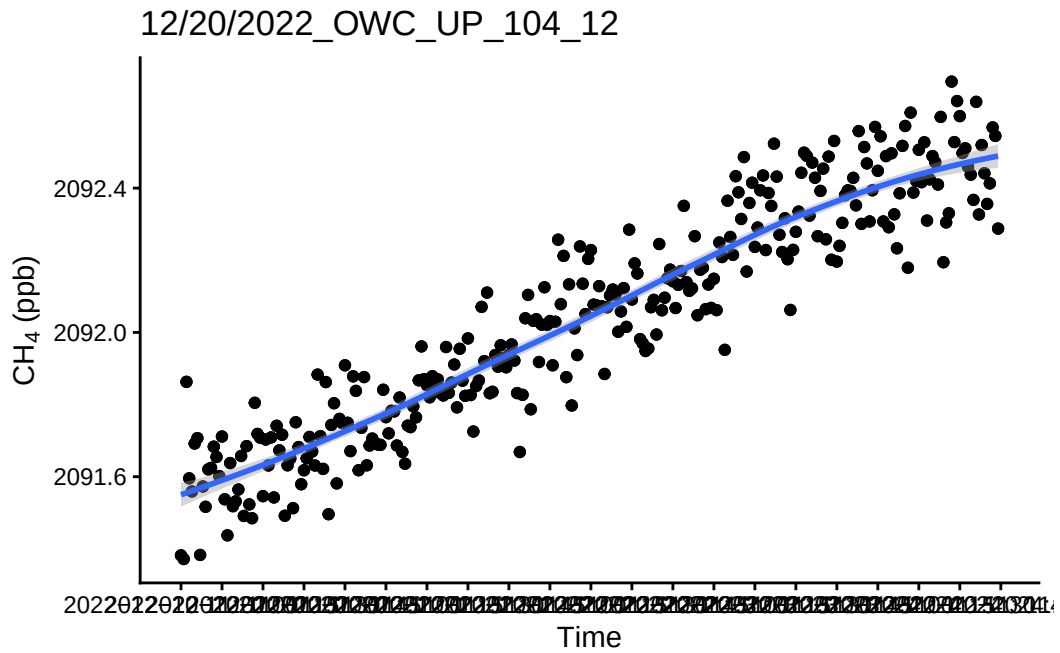
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



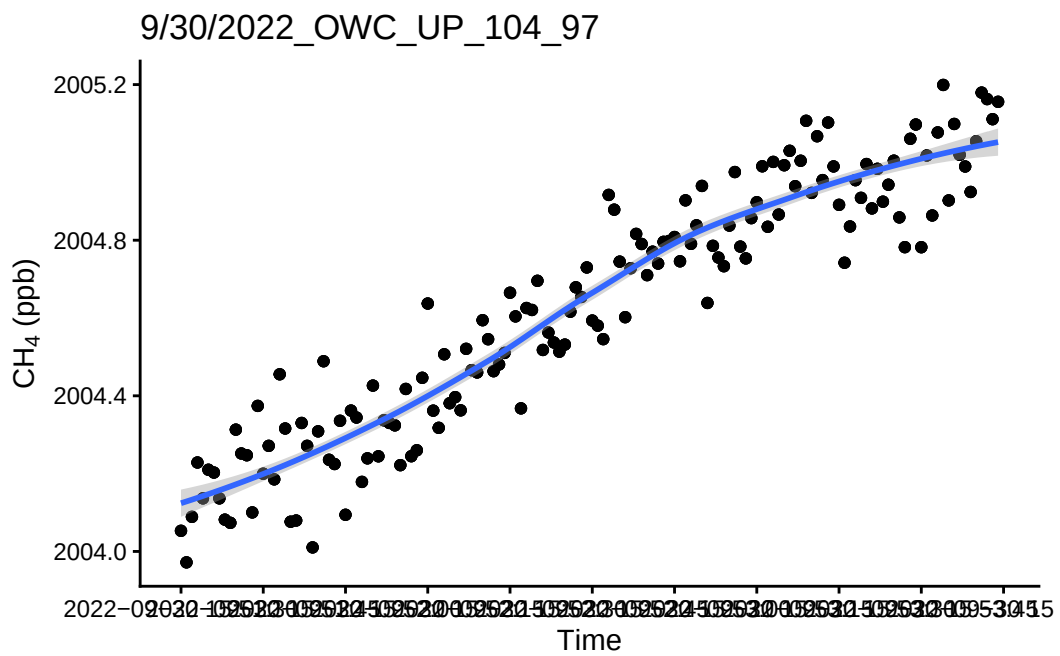
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



```
#####
```

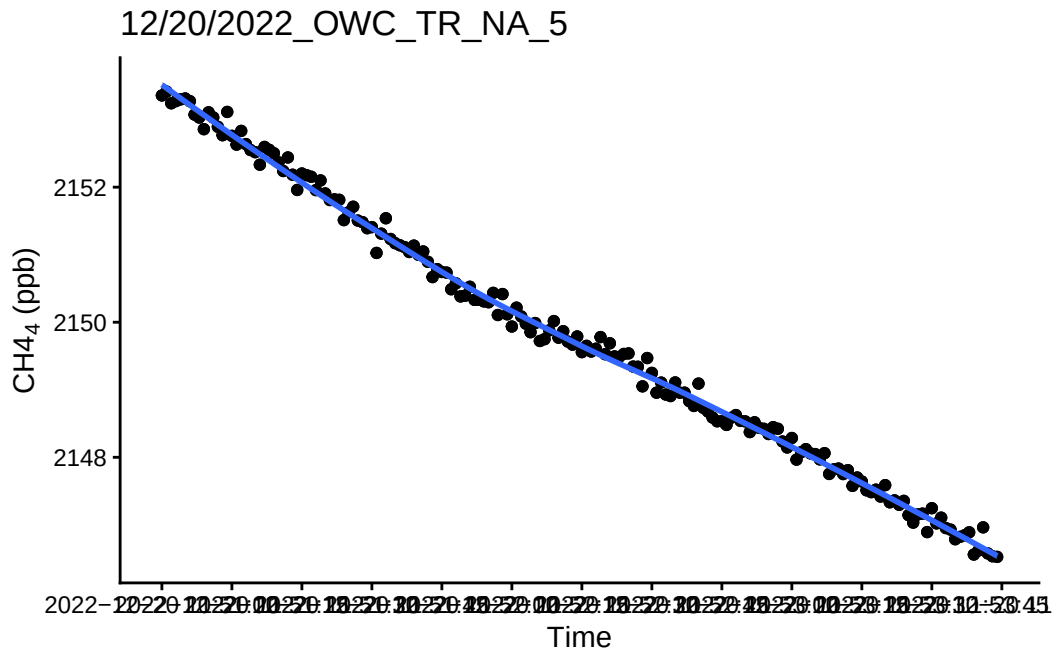
16 Look at the really high or really low ones to see if there is a timing issue or something

```
### pull out fluxes in the 5% and 95% tails to look at really high and really low
tail_plots <- unique(tails$Plot)

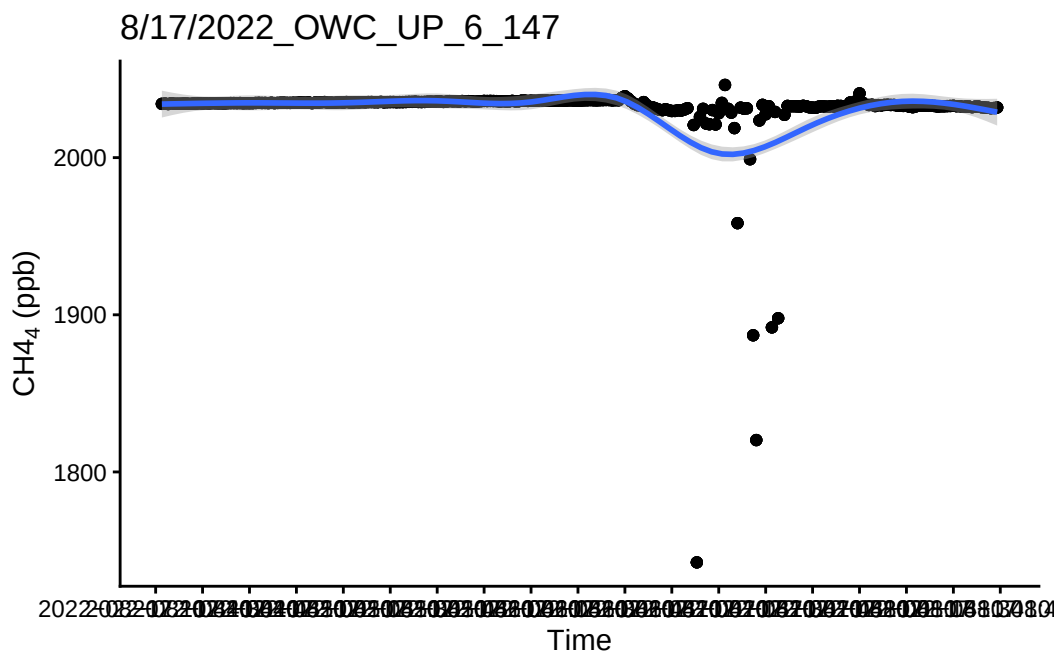
# take a random sample (FALSE so we don't get the same plot twice)
# set.seed(...) to guarantee reproducibility
tail_sample <- sample(tail_plots, 2, replace=FALSE)

#Loop to run five random plots subset above
for(P in tail_sample){
  r1 <- subset(alldat_trimmed, Plot == P & !is.na(metadata_row))
  p1 <- ggplot(r1, aes(TIMESTAMP, CH4)) + geom_point() + theme_classic() +
    scale_x_datetime(breaks = "15 sec") +
    ylab(expression(paste( CH4 [4], " (ppb)"))) +
    xlab("Time") + labs(title = P) +
    geom_smooth()
  print(p1)
}
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

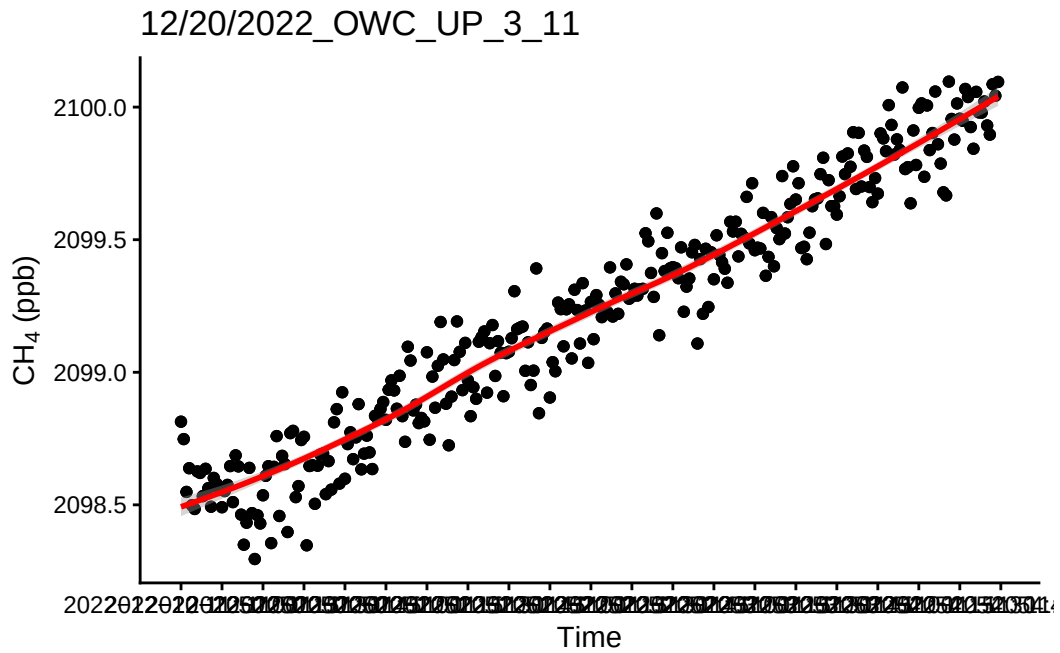


```
#would be nice to put on the plot the flux estimate and if it is a high or low or something  
#####
```

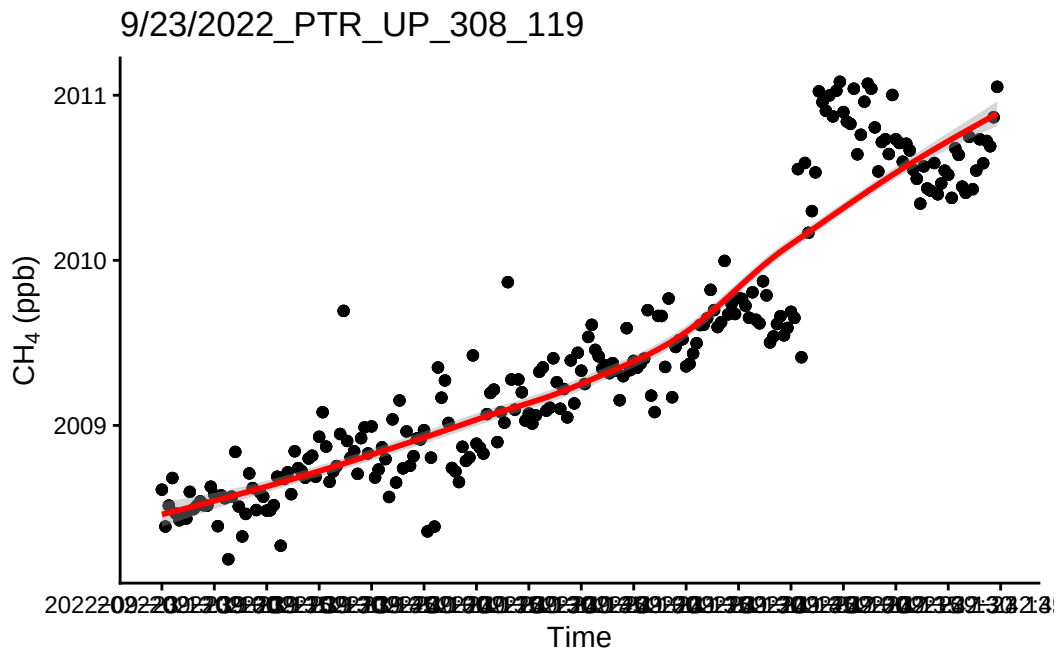
17 Check all fluxes

```
##Put all fluxes together  
fluxes_all <- as.data.frame(subset(fluxes))  
  
all_plots <- unique(fluxes_all$Plot)  
  
# take a random sample (FALSE so we don't get the same plot twice)  
# set.seed(...) to guarantee reproducibility  
all_sample <- sample(all_plots, 5, replace=FALSE)  
  
#Loop to run five random plots subset above  
for(P in all_sample){  
  r1 <- subset(alldat_trimmed, Plot== P & !is.na(metadat_row))  
  p1 <- ggplot(r1, aes(TIMESTAMP, CH4)) + geom_point() + theme_classic() +  
    scale_x_datetime(breaks = "15 sec") +  
    ylab(expression(paste( CH [4], " (ppb)"))) +  
    xlab("Time") + labs(title = P) +  
    geom_smooth(color="red")  
  print(p1)  
}
```

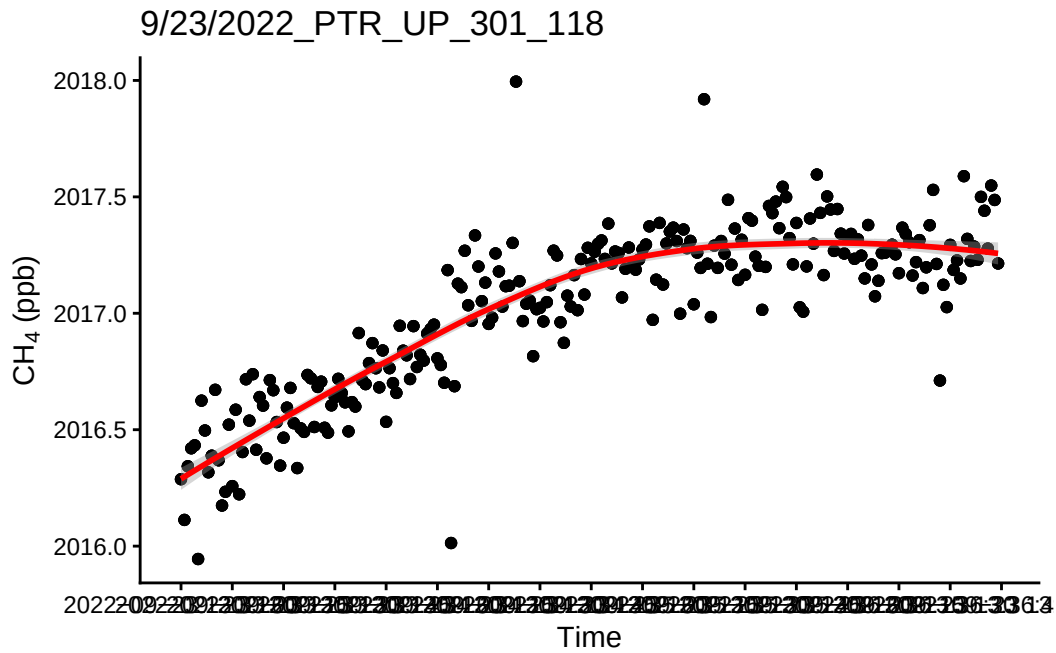
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



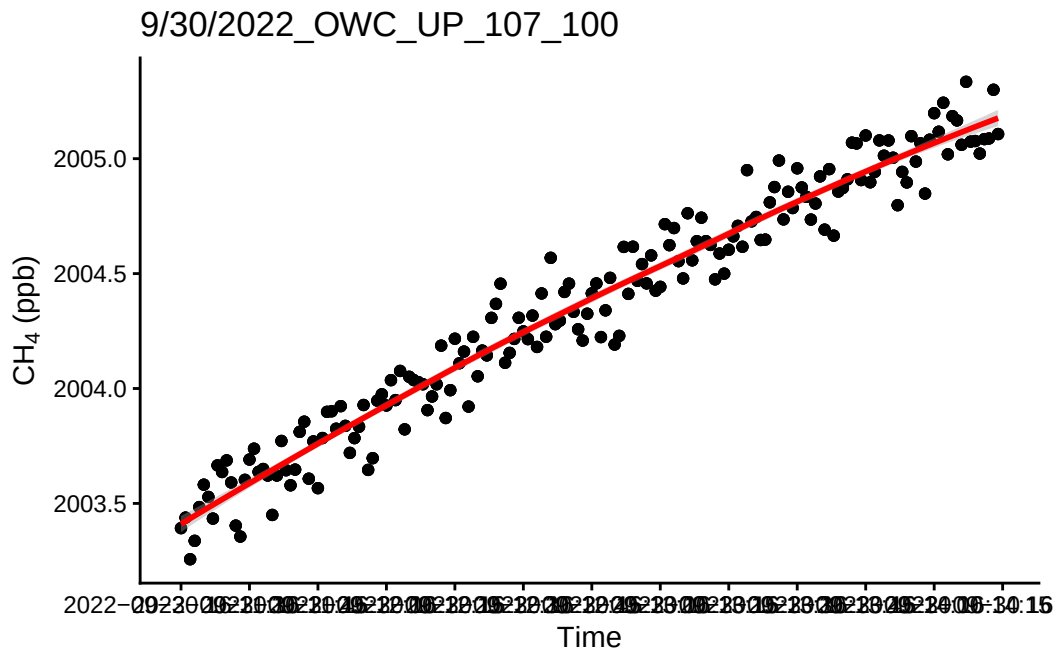
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



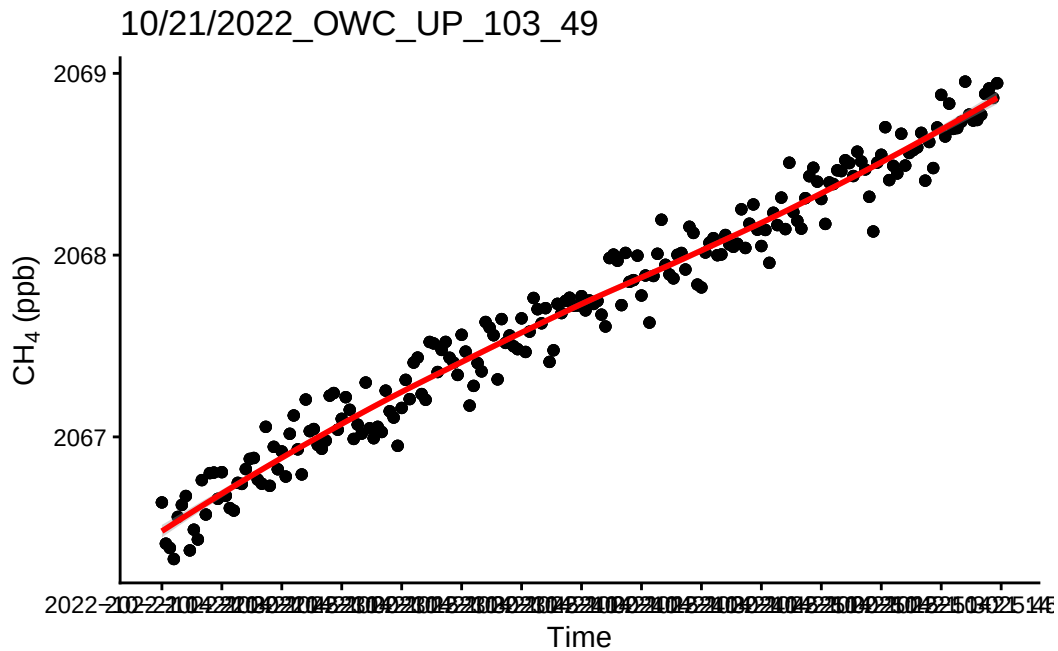
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



#####

18 QAQC flags

```
# QAQC flags
fluxes_flagged <- fluxes %>%
  mutate(
    # model fit flags
    CH4_lin_pass = (lin_r.squared >= 0.85 | lin_RMSE < 5) & !is.na(lin_r.squared),
    CH4_HM81_pass = (HM81_r.squared >= 0.85 | HM81_RMSE < 5) & !is.na(HM81_r.squared),
    CH4_poly_pass = (poly_r.squared >= 0.85 | poly_RMSE < 5) & !is.na(poly_r.squared),
    # negative flux flag (CH4 negative fluxes may be valid, but flagging anyway)
    CH4_neg_flux = lin_flux.estimate < 0 | is.na(lin_flux.estimate),
    # rejection flags
    CH4_rejected_model = !(CH4_lin_pass | CH4_HM81_pass | CH4_poly_pass)
  )

r2_summary <- fluxes_flagged %>%
  summarise(
    CH4_total_obs = n(),
```

```

CH4_n_rejected_model = sum(CH4_rejected_model, na.rm = TRUE),
CH4_pct_rejected_total = 100 * CH4_n_rejected_model / CH4_total_obs,
CH4_n_lin_pass       = sum(CH4_lin_pass, na.rm = TRUE),
CH4_n_HM81_pass      = sum(CH4_HM81_pass, na.rm = TRUE),
CH4_n_poly_pass      = sum(CH4_poly_pass, na.rm = TRUE)
)

# formatted table for PDF
r2_summary %>%
  tidyr::pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(
    Metric = recode(Metric,
      CH4_total_obs       = "Total observations",
      CH4_n_rejected_model = "Rejected - poor model fit (n)",
      CH4_pct_rejected_total = "Rejected - total (%)",
      CH4_n_lin_pass       = "Passed - Linear model (n)",
      CH4_n_HM81_pass      = "Passed - HM81 model (n)",
      CH4_n_poly_pass      = "Passed - Polynomial model (n)"
    ),
    Value = ifelse(Metric == "Rejected - total (%)",
      round(Value, 1),
      as.integer(Value))
  ) %>%
  kable(
    caption = "QA/QC CH4 Flux Summary",
    align   = c("l", "r"),
    booktabs = TRUE,
    format   = "latex"
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width    = FALSE
  ) %>%
  pack_rows("Rejections", 2, 3) %>%
  pack_rows("Model performance", 4, 6)

# R2 vs RMSE diagnostic plot
set.seed(42)
plot_data <- fluxes_flagged %>%
  slice_sample(n = min(15000, nrow(.))) %>%
  select(
    lin_r.squared, lin_RMSE, CH4_lin_pass,

```

Table 1: QA/QC CH₄ Flux Summary

Metric	Value
Total observations	151.0
Rejections	
Rejected — poor model fit (n)	5.0
Rejected — total (%)	3.3
Model performance	
Passed — Linear model (n)	144.0
Passed — HM81 model (n)	63.0
Passed — Polynomial model (n)	146.0

```

  HM81_r.squared, HM81_RMSE, CH4_HM81_pass,
  poly_r.squared, poly_RMSE, CH4_poly_pass
) %>%
rename(
  lin_pass = CH4_lin_pass,
  HM81_pass = CH4_HM81_pass,
  poly_pass = CH4_poly_pass
) %>%
pivot_longer(cols = everything(),
  names_to = c("model", ".value"),
  names_pattern = "(lin|HM81|poly)_(.*)")
) %>%
rename(r2 = r.squared) %>%
mutate(
  model = recode(model, lin = "Linear", HM81 = "HM81", poly = "Polynomial"),
  status = case_when(
    pass ~ "Accepted",
    TRUE ~ "Rejected"
  )
) %>%
filter(!is.na(r2), !is.na(RMSE))

QAQC_plot <- ggplot(plot_data, aes(x = r2, y = RMSE, color = model, shape = status)) +
  geom_point(size = 2) +
  geom_vline(xintercept = 0.85, linetype = "dashed", color = "black", linewidth = 0.8) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "black", linewidth = 0.8) +
  annotate("text", x = 0.7, y = max(plot_data$RMSE, na.rm = TRUE) * 0.95,
    label = "R2 = 0.85", hjust = -0.1, size = 3.5, fontface = "bold") +

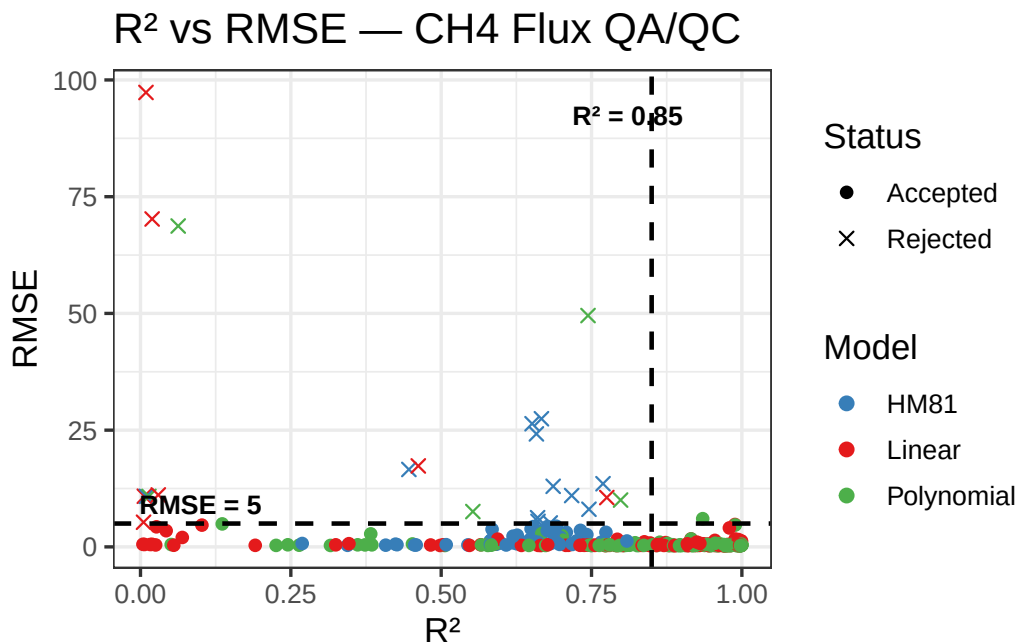
```

```

annotate("text", x = 0.1, y = 5,
        label = "RMSE = 5", vjust = -0.5, size = 3.5, fontface = "bold") +
scale_color_manual(values = c(
  "Linear"      = "#E41A1C",
  "HM81"       = "#377EB8",
  "Polynomial" = "#4DAF4A"
)) +
scale_shape_manual(values = c(
  "Accepted"    = 16,
  "Rejected"    = 4
)) +
labs(
  title = "R2 vs RMSE - CH4 Flux QA/QC",
  x     = "R2",
  y     = "RMSE",
  color = "Model",
  shape = "Status"
) +
theme_bw(base_size = 13) +
theme(legend.position = "right")

print(QAQC_plot)

```



19 Adding manual flag to reject any fluxes that do not fit criteria

```
# adding manual flags to the ones that are crazy off based on visual checks
manual_reject <- c()

fluxes_flagged <- fluxes_flagged %>%
  mutate(
    CH4_rejected_manual = Plot %in% manual_reject,
    CH4_flux_rejected   = CH4_rejected_model | CH4_rejected_manual)

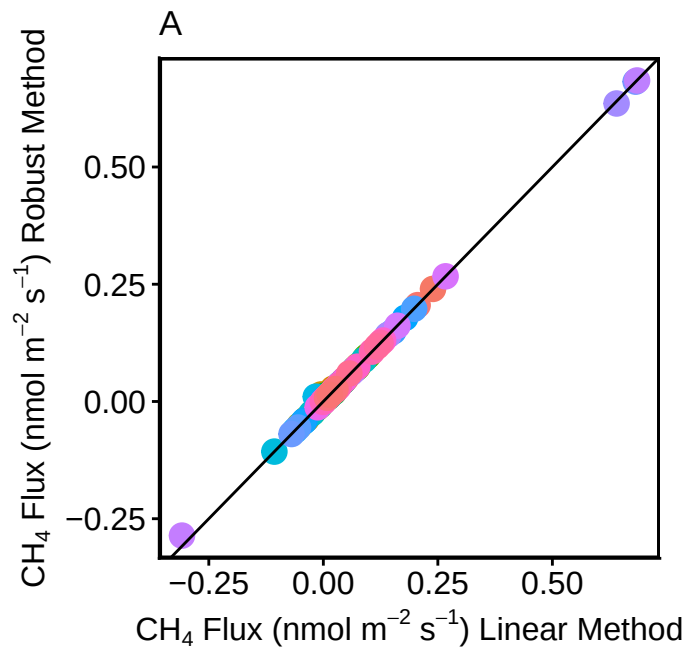
# fluxes_flagged %>%
#   filter(CO2_rejected_manual) %>%
#   select(Plot, lin_r.squared, lin_RMSE, CH4_rejected_model, CH4_rejected_manual, CH4_flux_rejected)
```

20 Do some visualization of QAQC plots to assess fluxes

```
allflux_filtered <- fluxes_flagged %>% filter(!CH4_rejected_model) #CHANGE FOR CH4_flux_rejected

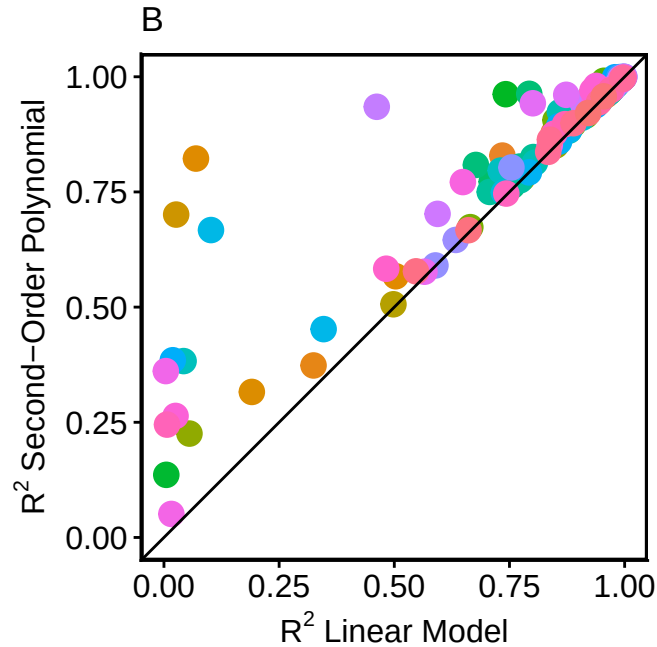
p1 <- ggplot(allflux_filtered, aes(lin_flux.estimate, rob_flux.estimate, color = Plot)) +
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  labs(title = "A") +
  ylab(expression(paste(CH [4], " Flux (nmol m-2 s-1) Robust Method"))) +
  xlab(expression(paste(CH [4], " Flux (nmol m-2 s-1) Linear Method"))) +
  guides(color = guide_legend(title = "Plot")) +
  theme(legend.position="none") +
  theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
        axis.title.y = element_text(size=12), legend.text=element_text(size=12),
        panel.border = element_rect(colour = "black", fill=NA, linewidth = 1),
        aspect.ratio = 1)

p1
```



```
p2 <- ggplot(allflux_filtered, aes(lin_r.squared, poly_r.squared, color = Plot)) +
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  xlim(0,1) + ylim(0,1) +
  labs(title = "B") +
  ylab(expression(paste("R"2* " Second-Order Polynomial")))) +
  xlab(expression(paste("R"2* " Linear Model")))) +
  guides(color = guide_legend(title = "Plot_ID")) +
  theme(legend.position="none") +
  theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
        axis.title.y = element_text(size=12), legend.text=element_text(size=12),
        panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
        aspect.ratio = 1)
```

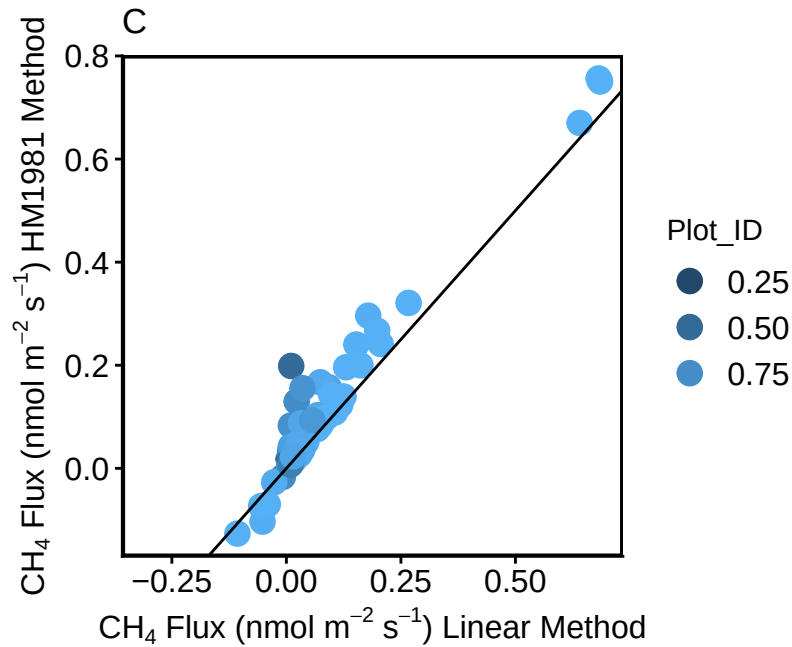
p2



```
p3 <- ggplot(allflux_filtered, aes(lin_flux.estimate, HM81_flux.estimate, color = lin_r.squa
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  #xlim(0,50) + #ylim(0,50) +
  labs(title = "C") +
  ylab(expression(paste( CH [4], " Flux (nmol m-2* " s-1*) HM1981 Method"))) +
  xlab(expression(paste( CH [4], " Flux (nmol m-2* " s-1*) Linear Method"))) +
  guides(color = guide_legend(title = "Plot_ID")) +
  theme(legend.position="right") +
  theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
        axis.title.y = element_text(size=12), legend.text=element_text(size=12),
        panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
        aspect.ratio = 1)
```

p3

Warning: Removed 73 rows containing missing values or values outside the scale range (``geom_point()``).



```
#ggarrange(p1, p2, p3, ncol=3, nrow=1, common.legend = TRUE, legend="none")
```

21 Read out final dataframe and export for use elsewhere

```
allflux_filtered_renamed <- fluxes_flagged%>%
  mutate(
    Date = mdy(str_extract(Plot, "[0-9]+/[0-9]+/[0-9]{4}")),
    Month = month.abb[month(Date)],
    Site = str_extract(Plot, "(PTR|OWC|CRC)"),
    Zone = str_extract(Plot, "_TR_|_UP_") %>%
      str_remove_all("_"),
    Replicate = as.integer(
      str_extract(Plot, "(?<=_ (TR|UP)_)[0-9]+")
    ),
    Row_ID = as.integer(
      str_extract(Plot, "[0-9]+$") ) %>%

# Optional: rebuild a clean Plot label
  mutate(
    Plot_clean = paste(
```

```

        format(Date, "%Y-%m-%d"),
        Site,
        Zone,
        Replicate,
        Row_ID,
        sep = "_"
    )
) %>%

select(
  Plot_clean,
  Date,
  Month,
  Site,
  Zone,
  Replicate,
  Row_ID,
  everything(),
  -Plot
)

#export as a csv to the folder that you want
write.csv(allflux_filtered_renamed, file="Processed Data/COMPASS_Synoptic_LE_TreeStem_Fluxes.csv")
# #Change file name

```

22